

Problem Set 4

The Definite Integral

September 14, 2026

Problem 1

(a) The drone is climbing from $t \in (0, 3)$. The drone is descending from $t \in (3, 5)$. It is highest at $t = 3$.

(b)

$$\overline{S}_n = 10m$$

$$\underline{S}_n = 0m$$

$\underline{S}_n$ uses right endpoints as the function is monotonically decreasing.

(c)

$$\int_0^5 v(t) dt = 9 - 4 = 5m$$

The drone's overall displacement in the interval is positive.

(d)

$$v(t) = 6 - 2t$$

$$R_n = \frac{5}{n} \sum_{k=1}^n v\left(\frac{5k}{n}\right) = 30 - \frac{25n(n+1)}{n^2} = 5 - \frac{25}{n}$$

$$\lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} 5 - \frac{25}{n} = 5$$

The limit is the same as the result in (c).

(e)

$$\int_0^5 |v(t)| dt = 9 + 4 = 13m$$

It is different from (c) as it considers the sign of the values. The displacement represents the drone's altitude, while distance is the log of meters flown.

(f)

$$M_5 = 5m$$

The midpoint sum is exact, as the left side of the midpoint balances with the right side of the midpoint.

Problem 2

(a)

$$\begin{aligned}
 x(x-2) = 0 &\implies x = 0, 2 \\
 y = (x-1)^2 - 1 &\implies x = 1 \\
 f(3) &= 3
 \end{aligned}$$

(b)

$$\begin{aligned}
 R_n &= \frac{3}{n} \sum_{k=1}^n f\left(\frac{3k}{n}\right) \\
 &= \frac{3}{n} \left(\frac{9}{n^2} \cdot \frac{n(n+1)(2n+1)}{6} - \frac{6}{n} \cdot \frac{n(n+1)}{2} \right) \\
 &= \frac{9(n+1)(2n+1)}{2n^2} - \frac{9(n+1)}{n} \\
 &= \frac{9(n+1)}{n} \left(\frac{2n+1}{2n} - \frac{2n}{2n} \right) \\
 &= \frac{9(n+1)}{2n^2} \\
 \int_0^3 (x^2 - 2x) dx &= \lim_{n \rightarrow \infty} R_n = 0
 \end{aligned}$$

(c)

$$\begin{aligned}
 R_n &= \frac{2}{n} \sum_{k=1}^n f\left(\frac{2k}{n}\right) \\
 &= \frac{2}{n} \left(\frac{4}{n^2} \cdot \frac{n(n+1)(2n+1)}{6} - \frac{4}{n} \cdot \frac{n(n+1)}{2} \right) \\
 &= \frac{4(2n^2 + 3n + 1)}{3n^2} - \frac{4(n+1)}{n} \\
 &= \frac{8}{3} + \frac{4}{n} + \frac{4}{3n^2} - 4 - \frac{4}{n} \\
 &= -\frac{4}{3} + \frac{4}{3n^2} \\
 \int_0^2 (x^2 - 2x) dx &= \lim_{n \rightarrow \infty} R_n = -\frac{4}{3} \\
 \int_2^3 (x^2 - 2x) dx &= \frac{4}{3} \\
 A &= \frac{8}{3}
 \end{aligned}$$

(d)

$$\int_0^3 (x-1)^2 dx = \int_0^3 x^2 - 2x + 1 dx = 0 + 3 = 3$$

$$\int_0^3 (x-1)^2 dx = \int_{-1}^2 x^2 dx = 3$$

Problem 5

(a)

$$\begin{aligned} a &= -1, b = 1 \\ f\left(\frac{k}{n}\right) &= 1 - \frac{k^2}{n} \\ R_n &= \frac{2}{n} \cdot \sum_{k=1}^n f\left(\frac{k}{n}\right) = \\ &= \frac{2}{n} \left(n - \frac{1}{n^2} \cdot \frac{n(n+1)(2n+1)}{6} \right) \\ &= 2 - \frac{2n^2 + 3n + 1}{3n^2} \\ &= \frac{4}{3} - \frac{1}{n} - \frac{1}{3n^2} = \boxed{\frac{4}{3}} \end{aligned}$$

(b)

$$\begin{aligned} \sqrt{1+x} &\geq \sqrt{1+x^2}, \quad x \in [0, 1] \\ \int_0^1 \sqrt{1+x} dx &\geq \int_0^1 \sqrt{1+x^2} dx \\ 1 &\leq \int_0^1 \sqrt{1+x} dx \leq \sqrt{2} \end{aligned}$$

(c)

$$\begin{aligned} g(x) &= -g(x) \implies \int_{-a}^a g(x) dx = 0 \\ g(x) &= g(-x) \implies \int_{-a}^a g(x) dx = 2 \int_0^a g(x) dx \\ \int_{-1}^1 x^3 \sqrt{1-x^2} dx &= 0 \\ \int_{-1}^1 (x^5 + 3\sqrt{1-x^2}) dx &= \frac{3\pi}{2} \end{aligned}$$

(d)

$$\begin{aligned} \pi \cdot 1 \cdot \frac{1}{2} &= \frac{\pi}{2} \\ \pi \cdot \frac{\pi}{2} \cdot \frac{1}{2} &= \frac{\pi^2}{4} \end{aligned}$$

$$\frac{\pi}{2} \leq I \leq \frac{\pi^2}{4}$$

$$W = \frac{\pi^2}{4} - \frac{\pi}{2} = 0.90$$

Problem 6

(a)

$$R_n = \sum_{k=1}^n \sqrt{x_k} \Delta x$$

$$x_k = \frac{k^2}{n^2}$$

$$\Delta x = \frac{2k-1}{n^2}$$

$$\frac{2n-1}{n^2} \rightarrow 0 \implies n \rightarrow \infty$$

$$R_n = \frac{1}{n^3} \sum_{k=1}^n (2k^2 - k)$$

$$= \frac{1}{n^3} \left(\frac{n(n+1)(2n+1)}{3} - \frac{n(n+1)}{2} \right)$$

$$= \frac{n+1}{n^2} \left(\frac{2n+1}{3} - \frac{1}{2} \right)$$

$$\lim_{n \rightarrow \infty} R_n = \frac{2}{3}$$

$$\int_0^1 \sqrt{x} \, dx = \frac{2}{3}$$

(b) They are reflections over $y = x$.

$$\int_0^1 x^2 \, dx = \frac{1}{3} x^3 \Big|_0^1 = \frac{1}{3}$$

$$1 - \frac{1}{3} = \frac{2}{3}$$

(c)

$$x_k = b \frac{k^2}{n^2}$$

$$\Delta x = b \frac{2k-1}{n^2}$$

$$\sqrt{x_k} = \sqrt{b} \frac{k}{n}$$

$$R_n = \sum_{k=1}^n \sqrt{x_k} \Delta x = \sum_{k=1}^n \left(\sqrt{b} \frac{k}{n} \right) \left(b \frac{2k-1}{n^2} \right)$$

$$\boxed{\int_0^b \sqrt{x} \, dx = \lim_{n \rightarrow \infty} R_n = \frac{2}{3} b^{\frac{3}{2}}}$$

(d)

$$\Delta x = \frac{1}{2(n-1)}$$

$$R_n = \sum_{k=1}^{n-1} \frac{k}{2n(n-1)} + p_n \cdot \frac{1}{2} = \frac{n+1}{8(n-1)} + p_n \cdot \frac{1}{2}, \quad p_n \in \left[\frac{1}{2}, 1\right]$$

$$\boxed{\lim_{n \rightarrow \infty} R_n = \frac{1}{8} + \frac{p_n}{2}}$$

The Δx for every single partition must go to zero. As the Δx for the last piece remains $\frac{1}{2}$ regardless of the value of n , the sums do not converge.